

Problem 53. Let n, m be positive integers such that $n \geq 2m + 1$. Given n distinct points and nm distinct line segments, each joining a pair of points, prove that there exist distinct points $A_0, A_1, \dots, A_m$, so that for each $i = 0, 1, \dots, m - 1$, there is a line segment joining A_i to A_{i+1} .

Solution. Fix m . We shall prove by induction on n . When $n = 2m + 1$, there is a line segment joining every pair of points. Thus the conclusion and the base case hold.

Now suppose that for some $k \geq 2m + 1$, the conclusion holds. Consider $k + 1$ points and $(k + 1)m$ line segments. If there is a point B which is joined to $\leq m$ line segments, then among the remaining k points, there are at least km line segments and the conclusion holds by induction hypothesis.

Now suppose that every point is joined to $\geq m + 1$ line segments. Pick any point to be A_0 . Pick any point that is joined to A_0 as A_i . Now A_i is joined to $\geq m$ points besides A_0 . Pick any of those points as A_2 . This process can be continued until we pick A_m . Thus the inductive step is established and we are done. $\square$

Problem 54. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(-f(x) - f(y)) = 1 - x - y$ for all $x, y \in \mathbb{Z}$.

Solution. The answer is $f(x) = x - 1$ for all $x \in \mathbb{Z}$.

One can check easily that $f(x) = x - 1$ satisfies the functional equation. Suppose $f: \mathbb{Z} \rightarrow \mathbb{Z}$ satisfies $f(-f(x) - f(y)) = 1 - x - y$. Substituting $y = 1$, we have $f(-f(x) - f(1)) = -x$. Thus, $f(-2f(1)) = -1$.

Let $n \in \mathbb{Z}$. Then $f(-f(n) - f(1)) = -n$. Letting $x = -f(n) - f(1)$ and $y = -2f(1)$, we get $f(n + 1) = 1 + (f(n) + f(1)) + 2f(1)$. That is $f(n + 1) = f(n) + 3f(1) + 1$. Using both forward and backward induction starting at $n = 1$, one can prove that

$$f(n) = (3n - 2)f(1) + n - 1.$$

Letting $n = 0$, we have $f(0) = -2f(1) - 1$. Substituting $x = 0$ and $y = 1$, we get $f(-f(0) - f(1)) = 0$, or equivalently, $f(f(1) + 1) = 0$. If we let $n = f(1) + 1$, we get $(3(f(1) + 1) - 2)f(1) + (f(1) + 1) - 1 = 0$. Simplifying, we have $f(1)(3f(1) + 2) = 0$. Since $f(1)$ is an integer, we must have $f(1) = 0$. Consequently, we have $f(n) = n - 1$ for all $n \in \mathbb{Z}$. $\square$

Problem 55. Let $n \geq 3$ be an odd integer. Suppose you play the following game using an $n \times n$ grid made up of n^2 unit squares. The game consists of n^2 turns. At each turn the following actions must be performed in the order specified:

- Choose an empty square and insert a positive integer greater than or equal to 1 and less than or equal to n^2 . Throughout the game, integers can be used only once.
- You gain 1 point if the sum of the integers written in the squares lying on the same row as the square you chose is a multiple of n . The same rule applies for columns. You get 2 points if both the row sum and the column sum are multiples of n .

Find the maximum possible value for the total sum of the points you gain by the end of the game.

Solution. The answer is $n(n + 1)$.

Due to the problem conditions we only need to consider the remainder of each positive integer inserted when divided by n . So we may assume that the positive integers inserted are $0, 1, 2, \dots, n - 1$, where each is inserted exactly n times.

Now we show a method of inserting the positive integers for which the total sum of points at the end of the game is $n(n + 1)$. We insert with the following rule: the positive integer inserted on the square positioned on i th row from the top of the grid and j th column from the left of the grid is the remainder of $i + j$ when divided by n . Observe that each of $0, 1, 2, \dots, n - 1$ appears exactly once in each row. Indeed, if in the i th row we have two entries, say $i + j$ and $i + k$ with $i + j \equiv i + k \pmod{n}$, then $j \equiv k \pmod{n}$, and so $j = k$ (because $1 \leq j, k \leq n$). With the same reasoning, each of $0, 1, 2, \dots, n - 1$ appears exactly once in each column. Hence the rule given above satisfies the problem conditions. Now we proceed with inserting the positive integers. First we insert 0 into the n designated squares to obtain $2n$ points, then we insert 1 and $n - 1$ into their designated squares to obtain $2n$ points, we insert 2 and $n - 2$ into their designated squares to obtain $2n$ points, $\dots$, and finally (because n is odd) we insert $\frac{n-1}{2}$ and $\frac{n+1}{2}$ into their designated squares to obtain $2n$ points. Hence total sum of points at the end of the game is

$$2n + 2n \cdot \frac{n-1}{2} = n(n+1).$$

Next we prove that the total sum of points at the end of the game is at most $n(n + 1)$. To this end, we define $A_0, A_1, \dots, A_{n^2}$ as follows: $A_0 = 0$, and

$$A_i = \begin{cases} A_{i-1} + 1 & \text{if the sum of the positive integers on the row of the } i\text{th turn (inclusive) is divisible by } n. \\ A_{i-1} & \text{otherwise,} \end{cases}$$

for $i = 1, 2, \dots, n^2$. We also define $B_0, B_1, \dots, B_{n^2}$ for columns in a similar way.

To find the maximum value of A_{n^2} , we define $C_0, C_1, \dots, C_{n^2}$ as follows: $C_0 = 0$ and C_i is the number of columns whose column sum after the i th turn is not divisible by n , for $i = 1, 2, \dots, n^2$. Now observe that

$$A_i + \frac{1}{2}C_i - \left(A_{i-1} + \frac{1}{2}C_{i-1}\right) \leq \begin{cases} 1 & \text{if 0 is inserted at the } i\text{th turn,} \\ \frac{1}{2} & \text{otherwise.} \end{cases}$$

Since 0 is inserted n times and the rest, namely $1, 2, \dots, n-1$, are inserted $n^2 - n = n(n-1)$ times, we have

$$A_{n^2} \leq A_{n^2} + \frac{1}{2}C_{n^2} \leq n + \frac{1}{2}n(n-1) = \frac{1}{2}n(n+1),$$

because (clearly), by definition, $C_{n^2} \geq 0$.

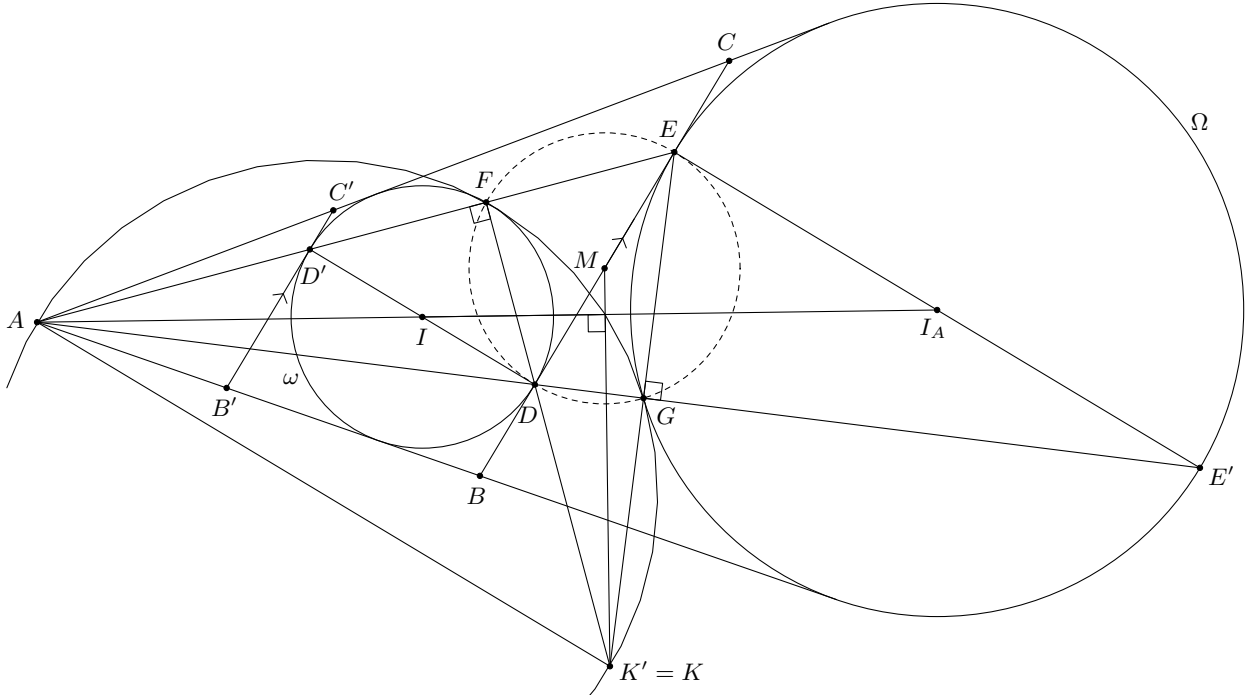
Similarly, we have $B_{n^2} \leq \frac{1}{2}n(n+1)$. Therefore total sum of points at the end of the game is

$$A_{n^2} + B_{n^2} \leq \frac{1}{2}n(n+1) + \frac{1}{2}n(n+1) = n(n+1). \quad \square$$

Problem 56. In $\triangle ABC$, let I be its incentre, ω be its incircle, M be the midpoint of BC , and K be the point of intersection of the line passing through A and perpendicular to line BC , and the line passing through M and perpendicular to the line AI .

Prove that the circle with AK as diameter is tangent to ω .

Solution 1.



If $AB = AC$, then M and K coincide and so the two circles are (clearly) tangent at M . Hence we may assume that $AB < AC$ as in the figure above.

Now we let

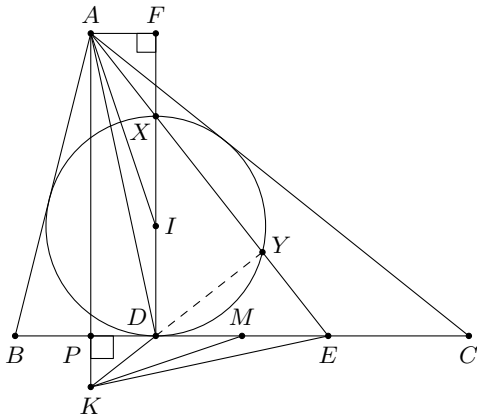
- $D = \omega \cap BC$;
- D' be the point on ω such that DD' is a diameter;
- Ω be the excircle opposite A with excentre I_A ;
- $E = \Omega \cap BC$;
- E' be the point on Ω such that EE' is a diameter;
- $F = \omega \cap AD'$ with $F \neq D'$;
- $G = \Omega \cap AE'$ with $G \neq E'$;
- $B' = AB \cap$ line through D' tangent to ω ;
- $C' = AC \cap$ line through D' tangent to ω ; and
- $K' = FD \cap EG$.

Then

- $B'C' \parallel BC$ (because both are perpendicular to DD') and so $\triangle AB'C' \sim \triangle ABC$;
- A, D', E are collinear and A, D, E' are collinear (because of the homothety centred at A);
- $\angle D'FD = 90^\circ = \angle DFE$ (because DD' is a diameter) and $\angle E'GE = 90^\circ = \angle EGD$ (because EE' is a diameter);
- F, D, G, E are concyclic (because $\angle DFE = 90^\circ = \angle EGD$);

- Therefore the circle with AK as diameter is tangent to ω at F (because of the homothety centred at F). $\square$

Solution 2.



Define the following points:

- D is the tangential point of BC with ω ;
- X is the point on ω such that DX is a diameter;
- E is the intersection of AX and BC ;
- P is the intersection of BC and AK ;
- F is the projection of A onto DI .

Recall that E is the tangential point of BC with the A -excircle of $\triangle ABC$. So M is the midpoint of DE . Since $KP \perp AF$, $PM \perp FI$ and $MK \perp IA$, we have $\triangle KPM \sim \triangle AFI$.

Since AF and DE are parallel, and $AFDP$ is a rectangle, we have $\frac{FX}{DX} = \frac{AF}{ED} = \frac{PD}{ED}$. As I and M are the midpoints of XD and DE respectively, this implies $FX : XI : ID = PD : DM : ME$. It follows from $\triangle AFI \sim \triangle KPM$ that $\triangle AFX \sim \triangle KPD$. Thus, $AX \perp KD$. This shows AX and KD meet at a point Y on ω (as XD is a diameter of ω).

Now, as AK and XD are parallel, there exists a homothety with centre Y that maps XD to AK . It maps ω to the circumcircle of $\triangle AKY$, which is the circle with diameter AK as $\angle AYK = 90^\circ$. The two circles must be tangent at Y by properties of homothety.